

CODEHIVE

Java Masterclass: Mathematics Logic & Operations

Introduction

The Math class is a utility class providing static methods for numeric operations. No instantiation is required.

```
// Accessing Math methods directly
double val = Math.sqrt(64); // result: 8.0
```

Extreme Values (Min/Max)

```
int high = Math.max(5, 10); // 10
int low  = Math.min(5, 10); // 5
```

Rounding & Absolute

```
Math.abs(-4.7);    // 4.7
Math.round(4.6);   // 5
Math.ceil(4.1);    // 5.0 (Rounds UP)
Math.floor(4.9);   // 4.0 (Rounds DOWN)
```

Powers & Randomness

```
Math.pow(2, 8);    // 256.0
double rnd = Math.random(); // 0.0 to 1.0

// Random integer (0-100)
int n = (int)(Math.random() * 101);
```

Logarithms & Trig

```
Math.log10(100); // 2.0
Math.exp(1);      // 2.718... (Euler's number)
Math.sin(Math.toRadians(90)); // 1.0
```

CodeHive Quick Summary

- **Static Access:** Call `Math.method()` directly.
- **Random:** Use `(int)(Math.random() * range) + min` for specific integers.
- **Pow/Sqrt:** Return double by default.
- **Rounding:** `ceil` is ceiling (up), `floor` is ground (down).